

(d) The chance of a horse jumping a fence without knocking it down is calculated as follows.

If the height of the fence is more than the maximum height a horse can jump, the success percentage is 20% of the horse's `PercentageSuccess`. The risk does not affect this value.

If the height of the fence is less than or equal to the maximum height a horse can jump, the risk gives a modifier value to multiply with the horse's `PercentageSuccess`.

The risk values and their modifiers are given in this table:

| Risk | Modifier |
|------|----------|
| 5    | 0.6      |
| 4    | 0.7      |
| 3    | 0.8      |
| 2    | 0.9      |
| 1    | 1.0      |

For example:

- The horse Jet has `PercentageSuccess` of 65 and `MaxFenceHeight` of 160.
- A fence has a height of 140 and a risk of 3.
- The height of the fence is less than the horse's `MaxFenceHeight`, therefore the risk is used.
- The risk of 3 gives the modifier 0.8.
- The modifier 0.8 is multiplied by the horse's `PercentageSuccess` of 65, which gives 52.
- The chance of the horse successfully jumping this fence is 52%.

The method `Success()` in the `Horse` class:

- takes the height and risk of a fence as parameters
- calculates the percentage chance of success for that horse jumping the fence without knocking it down
- returns the calculated percentage chance of success as a real number.

Write program code for `Success()`.

Save your program.

Copy and paste the program code into part 2(d) in the evidence document.

[5]